

Lab 5 - Malware Triage

Microsoft Defender Detection and Wazuh Correlation Using EICAR

Attribute	Details
Endpoint	Windows 11 VM
Security product	Microsoft Defender Antivirus
SIEM	Wazuh
Test artifact	EICAR antivirus test file
Exercise type	Authorized defensive validation

1. Objective

Perform safe malware-triage validation using the EICAR antivirus test file, observe Microsoft Defender detection and quarantine, and correlate the endpoint telemetry in Wazuh.

2. Safe Test Artifact

EICAR is an industry-standard harmless antivirus test string. It is designed to trigger security products without using real malware.

3. Detection Workflow

- Created the EICAR test file in a controlled lab directory.
- Observed Microsoft Defender detection and Protection History.
- Verified the file was quarantined.
- Collected Defender Operational events through the Wazuh agent.
- Investigated the detection and remediation events in the SIEM.

4. Key Evidence

Evidence	Observed value
Threat name	Virus:DOS/EICAR_Test_File
Defender Event ID	1116 - threat detected
Remediation Event ID	1117 - action taken
Wazuh detection rule	62123 - detection
Wazuh remediation rule	62124 - remediation
SHA-256	275A021BBFB6489E54D471899F7DB9D1663FC695EC2FE2A2 C4538AABF651FD0F

5. Analyst Interpretation

The antivirus detection was a true positive from the security product perspective because the EICAR test signature was intentionally present. However, the artifact was benign test content, not real malware. Defender successfully quarantined the file, and no further malicious behavior was observed.

6. Skills Demonstrated

- Microsoft Defender investigation
- Malware triage fundamentals
- Windows Defender Operational events
- Wazuh alert correlation
- Hash analysis
- Remediation verification
- Incident documentation

7. Safety and Scope

No real malware was executed. No antivirus exclusions or security-control bypasses were used. The exercise was limited to a controlled Windows VM and safe EICAR validation.

8. Conclusion

The lab confirmed that endpoint detection, Defender event logging, Wazuh ingestion, and remediation visibility were functioning correctly.